

KUNAL M SHENDE

Embedded Systems Engineer | IoT Security Researcher

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PROFESSIONAL SUMMARY

Computer Science undergraduate with two years of experience in embedded systems, hardware security, and web development. Proficient in vulnerability assessment, penetration testing, and firmware engineering. National competition award winner and Indian Design Patent holder.

TECHNICAL SKILLS

Programming Languages: C, Embedded C, C++, Python, TypeScript, Bash

Hardware Platforms: ESP32, ESP32-S3, Arduino, Raspberry Pi, FreeRTOS, SPI, I2C, UART

Web Development: HTML5, CSS3, Bootstrap, Python Flask, WebSockets, Firebase

Databases: MySQL, Supabase, Firebase Firestore

Security Tools: Burp Suite, Nmap, Wireshark, Metasploit, OWASP Top 10

DevOps and Cloud: Git, GitHub, Docker, Vercel, Google Cloud Platform, Visual Studio Code, Postman

WORK EXPERIENCE

Independent Security Researcher | Freelance | January 2024 - Present

- Architected three secure hardware prototypes for Internet of Things applications, reducing firmware attack surfaces by 40 percent.
- Identified 15 critical security vulnerabilities in 10 controlled test environments using industry standard assessment frameworks.
- Reduced network anomalies by 25 percent through packet-level traffic analysis and proactive system hardening measures.

Web Exploit Hunting Intern | EduSkills and AICTE | June 2026 - August 2026

- Detected 12 security flaws including injection and scripting vulnerabilities in eight web applications during testing exercises.
- Compiled detailed vulnerability reports with severity classifications, accelerating team remediation timelines by 30 percent.

Cybersecurity Intern | Palo Alto Networks and EduSkills | January 2026 - March 2026

- Reviewed 500 network packet captures and firewall configurations, strengthening endpoint protection for three cloud environments.
- Earned four industry certifications in security operations, network security, cybersecurity, and cloud security disciplines.

Java Full Stack Developer Virtual Intern | EduSkills and AICTE | October 2025 - December 2025

- Completed a structured 12-week training program in application development, database integration, and backend service design.

Cloud Virtual Intern | AWS Academy | April 2025 - June 2025

- Studied 10 core cloud infrastructure services over an eight-week hands-on laboratory curriculum covering compute and storage.

Android Developer Virtual Intern | Google for Developers | January 2025 - March 2025

- Explored eight mobile application development modules covering interface design, lifecycle management, and local data storage.

PROJECTS

ZeroTrace - Cybersecurity Audit Device | ESP32-S3, FreeRTOS, C++ | [\[GitHub Repo\]](#) | 2026

- Engineered an embedded security audit device integrating four modules for wireless scanning, signal analysis, and automated testing.
- Optimized graphical display rendering to 60 frames per second by distributing real-time tasks across dual processor cores.
- Secured second prize at a national technical competition among 50 participating teams for this project submission.

Sentinel - Attack Detection Framework | Python, Windows API | 2026

- Developed a real-time threat detection system that identifies and neutralizes rogue peripheral devices within 10 milliseconds.
- Validated the framework against simulated attack scenarios, achieving a 100 percent detection rate in controlled test environments.

MicroFlasher - Web Serial Flashing Platform | TypeScript, Web Serial API | 2026

- Deployed a browser-based firmware installation tool that eliminated setup requirements, reducing onboarding time by 70 percent.

Autonomous Line-Follower Robot | ATmega328P, L298N, IR Sensors | [\[GitHub Repo\]](#) | 2025

- Constructed a differential-drive robot with precision motor calibration, eliminating stalling and reducing noise output by 90 percent.
- Calibrated a proportional control algorithm that achieved 95 percent path tracking accuracy on sharp turn sections.

ACHIEVEMENTS

- Won second prize at the national level technical competition among 50 teams for the embedded security device project.
- Filed an Indian Design Patent in April 2026 for original hardware design work on the cybersecurity audit device.

EDUCATION

Bachelor of Technology in Computer Science and Engineering | S.B. Jain Institute of Technology, Nagpur | August 2023 - June 2027

Relevant Coursework: Data Structures, Computer Networks, Operating Systems, Embedded Systems, Cybersecurity

Higher Secondary Certificate in Science | Municipal Junior College, Kalmeshwar | June 2021 - March 2023

Secondary School Certificate with Distinction | Municipal Junior College, Kalmeshwar | June 2019 - March 2021

CERTIFICATIONS

Cyber Security Workshop - Crow Security and IIT Bombay, 2026

Ethical Hacking and Penetration Testing - NASSCOM FutureSkills Prime, 2026

Security Operations Fundamentals - Palo Alto Networks, March 2026

Network Security Fundamentals - Palo Alto Networks, March 2026

Cybersecurity Fundamentals - Palo Alto Networks, March 2026

Cloud Security Fundamentals - Palo Alto Networks, March 2026

Python Programming - HackerRank, 2025

Cloud Computing Fundamentals - AWS Academy, 2025

German Language Proficiency Level 1 - FLOA, December 2025

LANGUAGES

English (Professional) | Hindi (Native) | Marathi (Native) | German (Beginner)

INTERESTS

Embedded Systems Design, Hardware Security Research, Open Source Development, Competitive Robotics, Technical Writing